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Lab 1: Requirements Engineering & UML Use-Case Modelling

PROBLEM STATEMENT #26 | Smart Cities, Transport & Logistics

Peer-to-Peer Carpooling & Expense Splitter

Use-Case Flow Specification (UC-02 – Search Available Rides)

Primary Actor: Rider Commuter

Supporting Actor: Map/Route Service

Preconditions

- Driver Hosts have published rides with valid source, destination, travel time, and available-seat information.
- The Rider Commuter has access to the carpooling system.
- The Map/Route Service is available to provide route and distance information.

Postconditions

- The system displays rides that satisfy the Rider Commuter's search details and route-compatibility rules.
- Only rides requiring a driver detour of **1.5 km or less** are treated as compatible.
- The Rider Commuter may view the Driver Host's rating and verification status or proceed to request a suitable ride.

Main Success Scenario

1. The Rider Commuter selects **"Search Available Rides."**
2. The system prompts the Rider Commuter to enter the source, destination, and preferred travel time.
3. The Rider Commuter enters the required trip details and submits the search.
4. The system validates the entered information.
5. The system retrieves available rides published by Driver Hosts.
6. The system invokes **UC-03 – Calculate Route Compatibility** for the available rides.
7. The Map/Route Service provides the required route and distance information.
8. The system calculates the additional detour required for each Driver Host.
9. The system accepts only rides where the required detour is **1.5 km or less**.
10. The system displays the compatible rides to the Rider Commuter.
11. The Rider Commuter selects a suitable ride and may view the Driver Host's rating and verification status or proceed to **UC-05 – Request Ride**.
12. The use case ends successfully.

Alternate Flow – No Compatible Rides Found

9a. None of the available rides satisfy the **1.5 km detour tolerance**.

9a.1. The system displays **"No compatible rides found."**

9a.2. The system allows the Rider Commuter to modify the source, destination, or preferred travel time.

9a.3. The Rider Commuter modifies the search criteria and submits the search again.

9a.4. The system returns to Step 4 of the Main Success Scenario and performs the search again.